

Copula-Based Portfolio Risk Modeling Using S&P 500 Constituents

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1. Introduction and Objective

Traditional Value-at-Risk (VaR) estimation methods often rely on simplifying assumptions such as normally distributed returns or independence among asset returns. However, in real financial markets, asset returns exhibit non-linear dependence and tail co-movements that such models fail to capture.

This project proposes to develop a portfolio VaR estimation framework based on copula models using daily return data of S&P 500 constituent stocks. The key objective is to model the joint dependence structure accurately, particularly in the tails, and to evaluate whether copula-based VaR improves risk estimation over the assumption of independence. The performance of the model will be assessed via traffic light backtesting methods.

The specific steps include:

- Data collection and preprocessing of daily returns for S&P 500 stocks.
- Fitting marginal distributions to individual stock returns.
- Constructing elliptical copula models (Gaussian, Student-t) to capture dependencies.
- Simulating portfolio returns using the fitted copula and marginal distributions.
- Estimating VaR at different confidence levels from the simulated returns.
- Backtesting how many times actual losses exceed predicted VaR and evaluating model performance by comparing with benchmarks assuming independence among assets.

2. Methodology

2.1 Data Preparation

We downloaded stock prices from yahoo finance and transformed to lognormal return rates. Data ranges from 2010-01-01 to 2025-11-03. Missing log return values are filled with 0.

2.2 Marginal Distribution Modeling

2.3 Fitting Copulas to Data

2.3.1 Gaussian Copula Gaussian copula assumes that the correlation structure of multivariates is linear and can be captured by a multivariate normal distribution. The key parameter to estimate is the correlation matrix Σ .

We can either use the Pearson correlation matrix or the Kendall's tau matrix to estimate the correlation matrix for the Gaussian copula. However, the Kendall's tau matrix is more robust to outliers so we will use it in this project.

Assume we have n assets with historical log returns data. Let F_i be the empirical CDF of the log returns for asset i .

The steps to fit a Gaussian copula are as follows:

1. Compute the empirical Kendall's tau matrix from the empirical df of the log returns.

$$\tau_{ij} = P((F_i - F'_i)(F_j - F'_j) > 0) - P((F_i - F'_i)(F_j - F'_j) < 0)$$

Where (F'_i, F'_j) is an independent copy of (F_i, F_j) .

2. Convert the Kendall's tau matrix to the correlation matrix using the relation:

$$\rho_{ij} = \sin\left(\frac{\pi}{2}\tau_{ij}\right)$$

ρ_{ij} is the (i, j) -th entry of the correlation matrix Σ .

2.3.2 t Copula The Gaussian copula is simple but may not capture tail dependencies well. The t copula can model stronger symmetric tail dependence. In other words, t copula can capture heavier tails in both lower and upper tails.

The correlation matrix for the t copula is estimated in the same way as the Gaussian copula using the Kendall's tau matrix. The degrees of freedom parameter ν can be estimated using maximum likelihood estimation (MLE) or method of moments based on the empirical data.

The log-likelihood function for the t copula is given by:

$$\ln L(\nu; \Sigma, \hat{U}_1, \hat{U}_2, \dots, \hat{U}_n) = \sum_{t=1}^n \ln g_{\nu, \Sigma}(t_{\nu}^{-1}(\hat{U}_{t,1}), t_{\nu}^{-1}(\hat{U}_{t,2}), \dots, t_{\nu}^{-1}(\hat{U}_{t,d})) - \sum_{i=1}^d \sum_{t=1}^n \ln g_{\nu}(t_{\nu}^{-1}(\hat{U}_{t,i}))$$

Where, - $\hat{U}_t = (U_{t,1}, U_{t,2}, \dots, U_{t,d}) = (\hat{F}_1(X_{t,1}), \hat{F}_2(X_{t,2}), \dots, \hat{F}_d(X_{t,d}))$ are the pseudo-observations. - t_{ν}^{-1} is the inverse CDF of the univariate t distribution with ν degrees of freedom. - $g_{\nu, \Sigma}$ is the joint density of a random vector with $t_d(\nu, 0, \Sigma)$ distribution.

2.4 Simulate Paths

To estimate VaR using copula models, we will simulate multiple paths of portfolio returns based on the fitted copula and marginal distributions.

Assume we have n assets in the portfolio and we want to simulate N scenarios of portfolio returns over m days.

2.4.1 Gaussian Copula Steps to simulate portfolio returns using Gaussian copula:

1. Generate $ntimesm$ matrix of independent standard normal variables Z .
2. Apply Cholesky decomposition to the correlation matrix Σ obtained from the historical returns to get a lower triangular matrix L such that $\Sigma = LL^T$.
3. Transform the independent standard normal variables to correlated normal variables: $Y = ZL^T$.
4. Convert the correlated normal variables to uniform variables using the standard normal CDF: $U_{ij} = \Phi(Y_{ij})$ for $i = 1, \dots, n$ and $j = 1, \dots, m$.
5. Invert the uniform variables to the original marginal distributions using the inverse CDF (quantile function) of each asset's fitted marginal distribution: $X_{ij} = F_i^{-1}(U_{ij})$.
6. Repeat steps 1-5 for N simulations to obtain N scenarios of portfolio returns.

2.4.2 Student-t Copula Steps to simulate portfolio returns using Student-t copula:

1. Same as steps 1-3 in Gaussian copula.
2. Simulate $n \times m$ matrix of independent chi-squared variables W with ν degrees of freedom.
3. Scale the correlated normal variables to obtain Student-t variables: $T_{ij} = Y_{ij} / \sqrt{W_{ij} / \nu}$.
4. Convert the Student-t variables to uniform variables using the Student-t CDF: $U_{ij} = t_\nu(T_{ij})$ for $i = 1, \dots, n$ and $j = 1, \dots, m$.
5. Invert the uniform variables to the original marginal distributions using the inverse CDF (quantile function) of each asset's fitted marginal distribution: $X_{ij} = F_i^{-1}(U_{ij})$.
6. Repeat steps 1-5 for N simulations to obtain N scenarios of portfolio returns.

2.5 VaR Estimation

2.6 Benchmark and Backtesting

3. Results

Since the program involves intensive computation and takes a long time to run, we only refit asset marginal distributions and copula models every year and

simulate daily returns for the next year based on the fitted models. However, the VaR estimation and backtesting are done on a daily basis, in which we use a rolling window to calculate the VaR for the next few days (e.g., 10 days) and compare with the actual losses then count the number of exceedances over the testing period. Since the main focus is comparing the copula-based VaR with the independence assumption-based VaR rather than precise VaR values, this approximation is acceptable using yearly refitting.

In each refitting, we use the past 5 years of daily returns to fit the marginal distributions and copula models. The results below show the fitting results, simulated portfolio returns, VaR estimates, and backtesting results.

3.1 Distribution Fitting Results

3.2 Copula Fitting Results

The following are the copula fitting results at each refitting date:

```
fitting date: 2020-01-08 00:00:00, copula: t, nu: 12.417756424535327
fitting date: 2021-01-08 00:00:00, copula: t, nu: 9.195719667646452
fitting date: 2022-01-10 00:00:00, copula: t, nu: 9.117256117881958
fitting date: 2023-01-12 00:00:00, copula: t, nu: 10.180435891793604
fitting date: 2024-01-17 00:00:00, copula: t, nu: 10.288289682819688
fitting date: 2025-01-21 00:00:00, copula: t, nu: 11.462445359219407
```

From the results, we can see that the t copula is consistently selected as the best-fitting copula model at each refitting date, indicating the presence of tail dependence among the asset returns. The estimated degrees of freedom parameter ν ranges from approximately 7.96 to 9.82, suggesting moderate tail heaviness in the joint distribution of asset returns.

The correlation matrices are too large to show here in full, but here are the snippets of the correlation matrices at each refitting date. We can see that the correlations among assets do not change significantly over time.

```
fitting date: 2020-01-08 00:00:00, correlation matrix:
[[1.          0.25135828 0.22668861 ... 0.2755526  0.36510618 0.18096865]
 [0.25135828 1.          0.19294253 ... 0.22807553 0.32770435 0.11335533]
 [0.22668861 0.19294253 1.          ... 0.21777834 0.31088217 0.45800343]
 ...
 [0.2755526  0.22807553 0.21777834 ... 1.          0.3645142  0.15124268]
 [0.36510618 0.32770435 0.31088217 ... 0.3645142  1.          0.28515786]
 [0.18096865 0.11335533 0.45800343 ... 0.15124268 0.28515786 1.          ]]

fitting date: 2021-01-08 00:00:00, correlation matrix:
[[1.          0.26263353 0.21670038 ... 0.26932472 0.33672786 0.16073783]
 [0.26263353 1.          0.23304742 ... 0.29520966 0.41469945 0.16328013]
 [0.21670038 0.23304742 1.          ... 0.2171963  0.30843546 0.47078713]
 ...
```

```

[0.26932472 0.29520966 0.2171963 ... 1.          0.37279705 0.15226454]
[0.33672786 0.41469945 0.30843546 ... 0.37279705 1.          0.26804008]
[0.16073783 0.16328013 0.47078713 ... 0.15226454 0.26804008 1.          ]]

fitting date: 2022-01-10 00:00:00, correlation matrix:
[[1.          0.26520473 0.16530195 ... 0.2467958 0.26536886 0.12163457]
 [0.26520473 1.          0.24837493 ... 0.32711292 0.44485892 0.17722045]
 [0.16530195 0.24837493 1.          ... 0.17684218 0.25438244 0.45194714]
 ...
 [0.2467958 0.32711292 0.17684218 ... 1.          0.36129222 0.1092786 ]
 [0.26536886 0.44485892 0.25438244 ... 0.36129222 1.          0.21916796]
 [0.12163457 0.17722045 0.45194714 ... 0.1092786 0.21916796 1.          ]]

fitting date: 2023-01-12 00:00:00, correlation matrix:
[[1.          0.32303348 0.16946741 ... 0.29402492 0.2826605 0.15276508]
 [0.32303348 1.          0.30050247 ... 0.40269805 0.48868892 0.23283978]
 [0.16946741 0.30050247 1.          ... 0.20642493 0.27499455 0.47781056]
 ...
 [0.29402492 0.40269805 0.20642493 ... 1.          0.39341918 0.1329136 ]
 [0.2826605 0.48868892 0.27499455 ... 0.39341918 1.          0.24694012]
 [0.15276508 0.23283978 0.47781056 ... 0.1329136 0.24694012 1.          ]]

fitting date: 2024-01-17 00:00:00, correlation matrix:
[[1.          0.33219196 0.15426941 ... 0.27997159 0.27046316 0.13835175]
 [0.33219196 1.          0.30707339 ... 0.44573637 0.51498306 0.24933645]
 [0.15426941 0.30707339 1.          ... 0.18496381 0.26589756 0.46847979]
 ...
 [0.27997159 0.44573637 0.18496381 ... 1.          0.40843087 0.147931 ]
 [0.27046316 0.51498306 0.26589756 ... 0.40843087 1.          0.24424112]
 [0.13835175 0.24933645 0.46847979 ... 0.147931 0.24424112 1.          ]]

fitting date: 2025-01-21 00:00:00, correlation matrix:
[[1.          0.32144924 0.15336631 ... 0.27221015 0.25305857 0.14491005]
 [0.32144924 1.          0.30775105 ... 0.45122881 0.5131643 0.26961728]
 [0.15336631 0.30775105 1.          ... 0.17882451 0.25748707 0.47164354]
 ...
 [0.27221015 0.45122881 0.17882451 ... 1.          0.40739566 0.15282154]
 [0.25305857 0.5131643 0.25748707 ... 0.40739566 1.          0.24952815]
 [0.14491005 0.26961728 0.47164354 ... 0.15282154 0.24952815 1.          ]]

```

3.3 Simulated Portfolio Returns

Assume an equally weighted portfolio of all S&P 500 constituent stocks and calculate the 1% and 99% quantiles of the simulated portfolio returns at each refitting date. We can see that the copula-based simulations produce more

extreme quantiles compared to the independence assumption, reflecting the impact of dependencies among asset returns.

```
fitting date: 2020-01-08 00:00:00,  
  independent 1% quantile: -0.00037, copula 1% quantile: -0.01014,  
  independent 99% quantile: 0.00180, copula 99% quantile: 0.01154  
fitting date: 2021-01-08 00:00:00,  
  independent 1% quantile: -0.00065, copula 1% quantile: -0.01148,  
  independent 99% quantile: 0.00219, copula 99% quantile: 0.01308  
fitting date: 2022-01-10 00:00:00,  
  independent 1% quantile: -0.00051, copula 1% quantile: -0.01067,  
  independent 99% quantile: 0.00214, copula 99% quantile: 0.01234  
fitting date: 2023-01-12 00:00:00,  
  independent 1% quantile: -0.00061, copula 1% quantile: -0.01175,  
  independent 99% quantile: 0.00219, copula 99% quantile: 0.01335  
fitting date: 2024-01-17 00:00:00,  
  independent 1% quantile: -0.00067, copula 1% quantile: -0.01213,  
  independent 99% quantile: 0.00214, copula 99% quantile: 0.01363  
fitting date: 2025-01-21 00:00:00,  
  independent 1% quantile: -0.00063, copula 1% quantile: -0.01224,  
  independent 99% quantile: 0.00218, copula 99% quantile: 0.01378
```

3.4 VaR Estimates

3.5 Backtesting Results

Note

1. We referenced text book - “Quantitative Risk Management: Concepts, Techniques and Tools” by Alexander J. McNeil, Rüdiger Frey, and Paul Embrechts for the methodology of copula modeling and VaR estimation.
2. The code is in the zip file. We did not put code here because it is too long.